Solution Architecture

Team ID	LTVIP2026TMIDS36743
Project Name	visualization tool for electric vehicle charge and range analysis

Solution Architecture:

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** project analyzes EV battery performance and driving range using real-time and historical data. The architecture illustrates how data flows from the electric vehicle sensors to meaningful visual insights for users.

Flow:

EV Sensors → Data Collection Module → Application Logic (Python/Analytics Engine) → Database → Web UI (Dashboard) → Visualization & Insights

EV Sensors:

Collect real-time data such as battery charge level (SoC), voltage, current, temperature, speed, and energy consumption.

Data Collection Module:

Captures and transfers sensor data to the processing system in structured format.

Application Logic:

Processes raw data, performs range prediction calculations, analyzes energy consumption patterns, and generates insights.

Database:

Stores EV data including battery parameters, trip history, charging sessions, and performance metrics.

Web UI (Dashboard):

Provides an interactive platform where users can monitor battery status, analyze range predictions, and review energy trends.

Visualization & Insights:

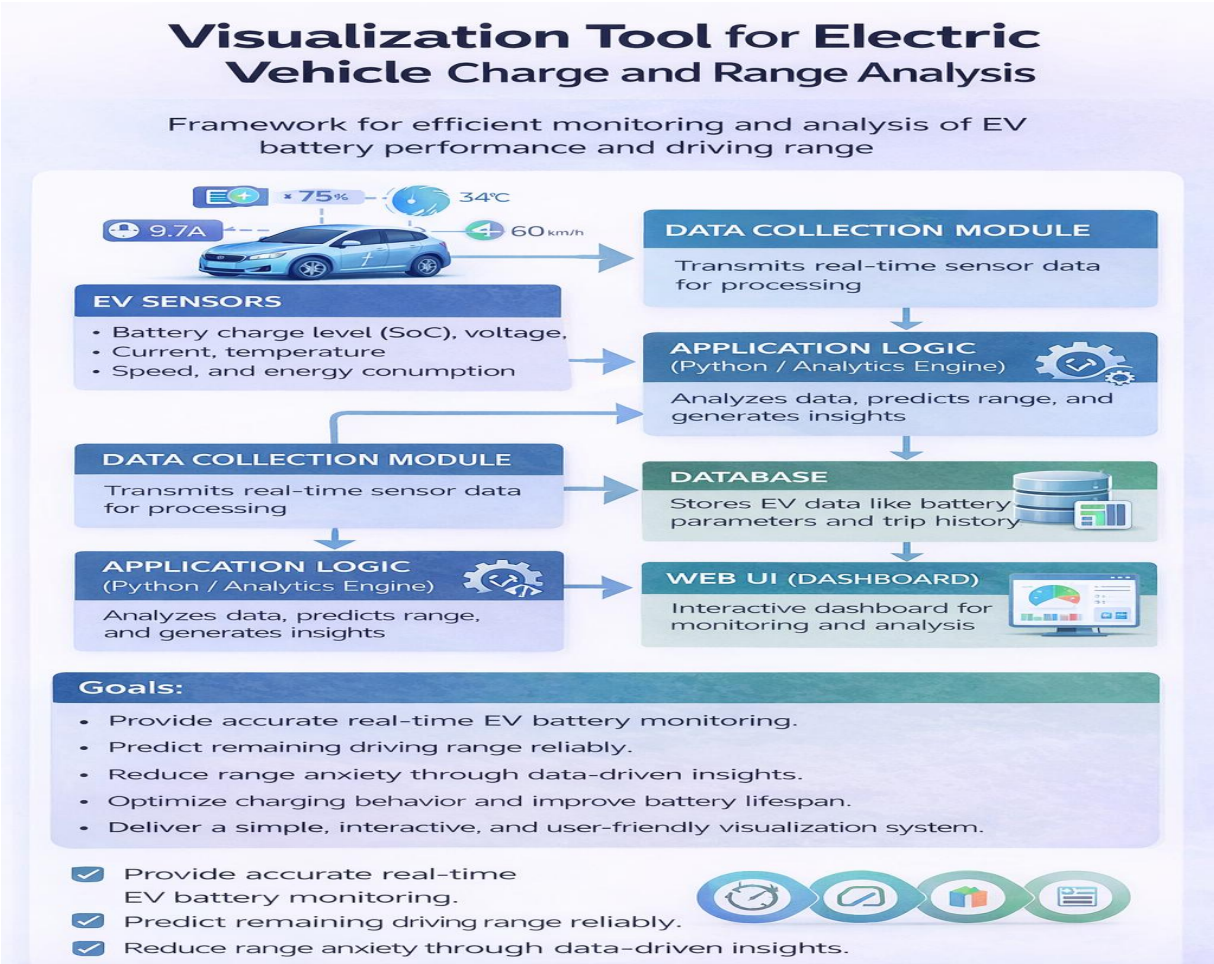
Displays charts, graphs, alerts, and comparative analysis to support better trip planning and charging decisions.

Goals:

Provide accurate real-time monitoring of EV battery performance.

Predict remaining driving range reliably.

Reduce range anxiety through data-driven insights.

Optimize charging behavior and improve battery lifespan.

Solution Architecture Diagram: